

Excessive Leucine-mTORC1-Signalling of Cow Milk-Based Infant Formula: The Missing Link to Understand Early Childhood Obesity

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Increased protein supply by feeding cow-milk-based infant formula in comparison to lower protein content of human milk is a well-recognized major risk factor of childhood obesity. However, there is yet no conclusive biochemical concept explaining the mechanisms of formula-induced childhood obesity. It is the intention of this article to provide the biochemical link between leucine-mediated signalling of mammalian milk proteins and adipogenesis as well as early adipogenic programming. Leucine has been identified as the predominant signal transducer of mammalian milk, which stimulates the nutrient-sensitive kinase mammalian target of rapamycin complex 1 (mTORC1). Leucine thus functions as a maternal-neonatal relay for mTORC1-dependent neonatal β -cell proliferation and insulin secretion. The mTORC1 target S6K1 plays a pivotal role in stimulation of mesenchymal stem cells to differentiate into adipocytes and to induce insulin resistance. It is of most critical concern that infant formulas provide higher amounts of leucine in comparison to human milk. Exaggerated leucine-mediated mTORC1-S6K1 signalling induced by infant formulas may thus explain increased adipogenesis and generation of lifelong elevated adipocyte numbers. Attenuation of mTORC1 signalling of infant formula by leucine restriction to physiologic lower levels of human milk offers a great chance for the prevention of childhood obesity and obesity-related metabolic diseases.